CURRICULUM SPECIFICATIONS
PHYSICS 1 (SP015)

1.	Name and code of course:	Physics 1 (SP015)
2.	Synopsis:	This course is offered to students to acquire concepts in physics as well as to enhance students' manipulative skills through practical sessions.
3.	Name(s) of academic staff:	Aifaa binti Awang Kechik
4.	Semester offered:	Semester I
5.	Credit value:	5
6.	Prerequisite/ co-requisite (if any):	None
7.	Course Learning Outcomes (CLO):	

At the end of the course, students should be able to:

1. Describe basic concepts of mechanics, waves, matters, heat and thermodynamics. (C2, PLO 1, MQF LOC i)

2. Solve problems related to mechanics, waves, matters, heat and thermodynamics. (C4, PLO 2, MQF LOC ii)

3. Apply the appropriate scientific laboratory skills in physics experiments. (P3, PLO 3, MQF LOC iii a)

Note: *The* ***MQF*** **2.0** ***learning outcomes clusters*** **(*LOC*)** *are*:

i. *Knowledge and understanding*
ii. *Cognitive skills*
iii. *Functional work skills with focus on:*
   a. *Practical skills*
   b. *Interpersonal skills*
   c. *Communication skills*
   d. *Digital skills*
   e. *Numeracy skills*
   f. *Leadership, autonomy and responsibility*
iv. *Personal and entrepreneurial skills*
v. *Ethics and professionalism*

8. Mapping of Programme Learning Outcomes (PLO) to Programme Educational Objectives (PEO)

PLO \ PEO	PEO 1	PEO 2	PEO 3	PEO 4	PEO 5
PLO 1	✓				
PLO 2		✓			
PLO 3	✓				
PLO 4			✓		
PLO 5			✓		
PLO 6				✓	
PLO 7		✓			
PLO 8					✓
PLO 9				✓	
PLO 10					✓

Programme Educational Objectives (PEO)

Upon a year of graduation from the programme, graduates are:

PEO 1 Knowledgeable and technically competent in science disciplines study in-line with higher educational institution requirement.
PEO 2 Able to apply information and use data to solve problems in science disciplines.
PEO 3 Able to communicate competently and collaborate effectively in group work to compete in higher education environment.
PEO 4 Able to use basic information technologies and engage in life-long learning to continue the acquisition of new knowledge and skills.
PEO 5 Able to demonstrate leadership skills and practice good values and ethics in managing organisations.

Programme Learning Outcomes (PLO)

At the end of the programme, students should be able to:

PLO 1 Acquire knowledge of science and mathematics as a fundamental of higher level education.
PLO 2 Apply logical, analytical and critical thinking in scientific studies and problem solving.
PLO 3 Demonstrate manipulative skills in laboratory works.
PLO 4 Collaborate in group work with skills required for higher education.
PLO 5 Deliver ideas, information, problems and solution in verbal and written communication.
PLO 6 Use basic digital technology to seek and analyse data for management of information.
PLO 7 Interpret familiar and uncomplicated numerical data to solve problems.
PLO 8 Demonstrate leadership, autonomy and responsibility in managing organization.
PLO 9 Initiate self-improvement through independent learning.
PLO 10 Practice good values attitude, ethics and accountability in STEM and professionalism.

9. Mapping of Course Learning Outcomes (CLO) to Programme Learning Outcomes (PLO), Teaching Methods and Assessment Methods:

CLO \ PLO	PLO 1	PLO 2	PLO 3	PLO 4	PLO 5	PLO 6	PLO 7	PLO 8	PLO 9	PLO 10	Teaching Method	Assessment Method
CLO 1	✓										Lecture	Summative Assessment Test
CLO 2		✓									Tutorial	Examination Assignment Lab Report
CLO 3			✓								Practical	Practical Test

(This description must be read together with Standard 2.1.2, 2.2.1 and 2.2.2 in Area 2)

10. Transferable Skills (if applicable):	Critical Thinking and Problem Solving

11. Distribution of Student Learning Time (SLT):

Course Content Outline	CLO	Learning & Teaching Activities: Guided Learning e-Learning (F2F)				Learning & Teaching Activities	Total SLT
		L	T	P	O	Independent Learning (NF2F)	
1.0 Physical Quantities and Measurements 1.1 Dimensions of physical quantities 1.2 Scalars and vectors 1.3 Significant figures and uncertainties analysis	1, 2, 3	1	3.5	2	0	4.5	11
2.0 Kinematics of Linear Motion 2.1 Linear motion 2.2 Uniformly accelerated motion 2.3 Projectile motion	1, 2, 3	1.5	8	2	0	9.5	21
3.0 Dynamics of Linear Motion 3.1 Momentum and impulse 3.2 Conservation of linear momentum 3.3 Basic of forces and free body diagram 3.4 Newton's Laws of Motion	1, 2	1.5	8	0	0	9.5	19
4.0 Work, Energy and Power 4.1 Work 4.2 Energy and Conservation of Energy 4.3 Power	1, 2, 3	1.5	7	2	0	8.5	19
5.0 Circular Motion 5.1 Parameters in circular motion 5.2 Uniform circular motion 5.3 Centripetal force	1, 2	1	2.5	0	0	3.5	7

	6.0 Rotation of Rigid Body 6.1 Rotational kinematics 6.2 Equilibrium of a uniform rigid body 6.3 Rotational dynamics 6.4 Conservation of angular momentum	1, 2, 3	1.5	7	2	0	8.5	19
	7.0 Simple Harmonic Motion and Waves 7.1 Kinematics of simple harmonic motion 7.2 Graphs of simple harmonic motion 7.3 Period of simple harmonic motion 7.4 Properties of waves 7.5 Superposition of waves 7.6 Application of standing waves 7.7 Doppler Effect	1,2,3	3.5	18	4	0	21.5	47
	8.0 Physics of Matters 8.1 Stress and strain 8.2 Young's Modulus 8.3 Heat conduction 8.4 Thermal expansion	1, 2	3.5	5	0	0	8.5	17
	9.0 Kinetic Theory of Gases and Thermodynamics 9.1 Kinetic theory of gases 9.2 Molecular kinetic energy and internal energy 9.3 First Law of Thermodynamics 9.4 Thermodynamic Processes 9.5 Thermodynamic Work	1, 2	3	7	0	0	10	20
	Total		**18**	**66**	**12**	**0**	**84**	**180**

Continuous Assessment	Percentage (%)	F2F	NF2F	Total SLT
1. Assignment (Topic 6)	10	0	3	3
2. Practical Test	15	1	3	4
3. Lab Report	15	1	3	4
Final Assessment	**Percentage (%)**	**F2F**	**NF2F**	**Total SLT**
1. Summative Assessment Test	20	1.5	4.5	6
2. Examination	40	2	6	8
GRAND TOTAL SLT				**205**

L = Lecture, T = Tutorial, P = Practical, O = Others, F2F = Face-to-Face, NF2F = Non Face-to-Face

12.	Special requirement or resources:	None

13.	References:	Cutnell, J. D., Johnson, K. W., Young, D., & Stadler, S. (2019). *Introduction to Physics* (11th ed.). Hoboken, N.J: John Wiley & Sons. Serway, R. A. & Jewett, J. A. (2019). *Physics for Scientists and Engineers* (10th ed.). International Student Edition. USA: Brooks/ Cole Cengage Learning. Giancoli, D. C. (2016). *Physics – Principles with Application* (7th ed.). Prentice Hall.
14.	Other additional information:	Teaching Learning Outcomes:

Topic 1: Physical Quantities and Measurements

1.1 Dimensions of physical quantities
- a) Define dimension.
- b) Determine the dimensions of derived quantities.
- c) Verify the homogeneity of equations using dimensional analysis.

1.2 Scalars and vectors
- a) Define scalar and vector quantities.
- b) Resolve vector into two perpendicular components (x and y axes).
- c) Determine resultant of vectors. (remarks: limit to three vectors only).

1.3 Significant figures and uncertainties analysis (Laboratory works)
- a) State the significant figures of a given number.
- b) Use the rules for stating the significant figures at the end of a calculation (addition, subtraction, multiplication or division).
- c) Determine the uncertainty for average value and derived quantities.
- d) Calculate basic combination (propagation) of uncertainties.
- e) State the sources of uncertainty in the results of an experiment.
- f) Draw a linear graph and determine its gradient, y-intercept and its respective uncertainties. (remarks: using Least Square Method LSM to determine uncertainties)
- g) Measure and determine the uncertainty of physical quantities. (Experiment 1: Measurement and uncertainty)

Topic 2: Kinematics of linear motion

2.1 Linear motion
- a) Define:
  - i. instantaneous velocity, average velocity and uniform velocity; and
  - ii. instantaneous acceleration, average acceleration and uniform acceleration.
- b) Interpret the physical meaning of displacement-time, velocity-time and acceleration-time graphs.
- c) Determine the distance travelled, displacement, velocity and acceleration from appropriate graphs.

2.2 Uniformly accelerated motion
- a) Derive and apply equations of motion with uniform acceleration

$$v = u + at\ ;\ v^2 = u^2 + 2as\ ;\ s = ut + \frac{1}{2}at^2\ ;\ s = \frac{1}{2}(u + v)t$$

2.3 Projectile motion
   a) Describe projectile motion launched at an angle, θ as well as special cases when θ=0°
   b) Solve problems related to projectile motion.
   c) Determine the acceleration due to gravity, g using free fall and projectile motion. (Experiment 2: Free fall and projectile motion)

Topic 3: Dynamics of Linear Motion

3.1 Momentum and impulse
   a) Define momentum and impulse, $J = F\Delta t$
   b) Solve problem related to impulse and impulse-momentum theorem, $J = \Delta p = mv - mu$
      **1D only*
   c) Use F-t graph to determine impulse.

3.2 Conservation of linear momentum
   a) State the principle of conservation of linear momentum.
   b) Apply the principle of conservation of momentum in elastic and inelastic collisions in 2D collisions.
   c) Differentiate elastic and inelastic collisions. (remarks: similarities & differences)

3.3 Basic of forces and free body diagram
   a) Identify the forces acting on a body in different situations:
      i. Weight, *W;*
      ii. Tension, *T;*
      iii. Normal force, *N;*
      iv. Friction, *f;* and
      v. External force (pull or push), *F.*
   b) Sketch free body diagram.
   c) Determine static and kinetic friction, $f_s \leq \mu_s N$, $f_k = \mu_k N$

3.4 Newton's Laws of Motion
   a) State Newton's laws of motion.
   b) Apply Newton's laws of motion.
      **include static and dynamic equilibrium for Newton's first law motion*

Topic 4: Work, Energy and Power

4.1 Work
- a) State the physical meaning of dot (scalar) product for work : $W = \vec{F} \bullet \vec{s} = Fscos\theta$
- b) Define and apply work done by a constant force.
- c) Determine work done from a force-displacement graph.

4.2 Energy and Conservation of Energy
- a) Define and use:
  - i. Gravitational potential energy, $U = mgh$
  - ii. Elastic potential energy for spring, $U_s = \frac{1}{2}kx^2 = \frac{1}{2}Fx$
  - iii. Kinetic energy, $K = \frac{1}{2}mv^2$
- b) State the principle of conservation of energy.
- c) Apply the principle of conservation of mechanical energy.
- d) State and apply work-energy theorem, $W = \Delta K$

4.3 Power
- a) Define and use average power, $P_{av} = \frac{\Delta W}{\Delta t}$ and instantaneous power, $P = \vec{F} \bullet \vec{v}$
- b) Verify the law of conservation of energy.
  (Experiment 3: Energy)

Topic 5: Circular Motion

5.1 Parameters in circular motion
- a) Define and use:
  - i. angular displacement, θ
  - ii. period, T
  - iii. frequency, f
  - iv. angular velocity, ω

5.2 Uniform circular motion
- a) Describe uniform circular motion.
- b) Convert units between degrees, radian, and revolution or rotation.

5.3 Centripetal force
- a) Explain centripetal acceleration and centripetal force, $a_c = \frac{v^2}{r} = r\omega^2 = v\omega$ and $F_c = \frac{mv^2}{r} = mr\omega^2 = mv\omega$
- b) Solve problems related to centripetal force for uniform circular motion cases: horizontal circular motion, vertical circular motion and conical pendulum.
  **exclude banked curve*

Topic 6: Rotation of Rigid Body

6.1 Rotational kinematics

a) Define and use:
   i. angular displacement, θ;
   ii. average angular velocity, ω_{av};
   iii. instantaneous angular velocity, ω;
   iv. average angular acceleration, α_{av}; and
   v. instantaneous angular acceleration, α.

b) Analyse parameters in rotational motion with their corresponding quantities in linear motion: $s = r\theta$, $v = r\omega$, $a_t = r\alpha$, $a_c = r\omega^2 = \frac{v^2}{r}$

c) Solve problem related to rotational motion with constant angular acceleration: $\omega = \omega_o + \alpha t$, $\theta = \omega_o + \frac{1}{2}\alpha t^2$, $\omega^2 = {\omega_o}^2 + 2\alpha\theta$, $\theta = \frac{1}{2}(\omega_o + \omega)t$

6.2 Equilibrium of a uniform rigid body

a) State the physical meaning of cross (vector) product for torque, $|\vec{\tau}| = rFsin\theta$

b) Define and apply torque.

c) State conditions for equilibrium of rigid body, $\sum F = 0$, $\sum \tau = 0$

d) Solve problems related to equilibrium of a uniform rigid body.
   **limit to 5 forces*

6.3 Rotational dynamics

a) Define and use moment of inertia, $I = \sum mr^2$

b) Use the moment of inertia of a uniform rigid body.
   (sphere, cylinder, ring, disc, and rod).

c) Determine the moment of inertia of a flywheel.
   (Experiment 4: Rotational motion of rigid body)

d) State and use net torque, $\sum \tau = I\alpha$

6.4 Conservation of angular momentum

a) Explain and use angular momentum, $L = I\omega$

b) State and use principle of conservation of angular momentum.

Topic 7: Simple Harmonic motion and waves

7.1 Kinematics of simple harmonic motion
a) Explain SHM.
b) Apply SHM displacement equation, $y = A\sin\omega t$
c) Derive and use equations:
i. velocity, $v = \omega A cos\omega t = \pm\omega\sqrt{A^2 - y^2}$
ii. acceleration, $a = -\omega^2 A\sin\omega t = -\omega^2 y$
(remarks: No calculus. Derive use algebra and trigonometry method, refer reference book Cutnell)
iii. kinetic energy, $K = \frac{1}{2}m\omega^2(A^2 - y^2)$
iv. potential energy, $U = \frac{1}{2}m\omega^2 y^2$
d) Emphasise the relationship between total SHM energy and amplitude.
e) Apply equations of velocity, acceleration, kinetic energy and potential energy for SHM.

7.2 Graphs of simple harmonic motion
a) Analyse the following graphs:
i. displacement-time;
ii. velocity-time;
iii. acceleration-time; and
iv. energy-displacement.

7.3 Period of simple harmonic motion
a) Use expression for period of SHM, *T* for simple pendulum and mass-spring system.
Simple pendulum : $T = 2\pi\sqrt{\frac{l}{g}}$, mass-spring system : $T = 2\pi\sqrt{\frac{m}{k}}$
b) Determine the acceleration, *g* due to gravity using simple pendulum.
(Experiment 5: SHM)
c) Investigate the effect of large amplitude oscillation to the accuracy of acceleration due to gravity, *g* obtained from the experiment.
(Experiment 5: SHM)

7.4 Properties of waves
a) Define wavelength.
b) Define and use wave number, $k = \frac{2\pi}{\lambda}$
c) Solve problems related to equation of progressive wave, $y(x,t) = A\sin(\omega t \pm kx)$
d) Distinguish between particle vibrational velocity and wave propagation velocity.
e) Use particle vibrational velocity, $v_y = A\omega\cos(\omega t \pm kx)$
f) Use wave propagation velocity, $v = f\lambda$
g) Analyse the graphs of:
i. displacement–time, *y-t*
ii. displacement–distance, *y-x*

7.5 Superposition of waves
a) State the principle of superposition of waves for the constructive and destructive interferences.
b) Use the standing wave equation,$y = 2A\ coskx\ sin\omega t$
c) Compare between progressive waves and standing waves.

7.6 Application of standing waves

a) Solve problems related to the fundamental and overtone frequencies for:
   i. stretched string, $f_n = \frac{nv}{2L}$ and
   ii. air columns (open, $f_n = \frac{nv}{2L}$ and closed end, $f_n = \frac{nv}{4L}$).

b) Use wave speed in a stretched string, $v = \sqrt{\frac{T}{\mu}}$

c) Investigate standing wave formed in a stretched string.
   (Experiment 6: Standing waves)

d) Determine the mass per unit length of the string.
   (Experiment 6: Standing waves)

7.7 Doppler Effect

a) State Doppler Effect for sound waves.

b) Apply Doppler Effect equation $f_a = \left(\frac{v \pm v_o}{v \mp v_s}\right) f$ for relative motion between source and observer. Limit to stationary observer and moving source, and vice versa.

Topic 8: Physics of Matters

8.1 Stress and strain

a) Distinguish between stress, $\sigma = \frac{F}{A}$ and strain, $\varepsilon = \frac{\Delta L}{L_o}$ for tensile and compression force.

b) Analyse the graph of stress-strain, $\sigma - \varepsilon$ for a metal under tension.

c) Explain elastic and plastic deformations.

d) Analyse graph of force–elongation, $F - \Delta L$ for brittle and ductile materials.

8.2 Young's Modulus

a) Define and use Young's Modulus, $Y = \frac{\sigma}{\varepsilon}$.

b) Apply strain energy, $U = \frac{1}{2} F \Delta L$ from force-elongation graph.

c) Apply strain energy per unit volume, $\frac{U}{V} = \frac{1}{2} \sigma \varepsilon$ from stress-strain graph.

8.3 Heat conduction

a) Define heat conduction.

b) Solve problems related to rate of heat transfer, $\frac{Q}{t} = -kA\left(\frac{\Delta T}{L}\right)$ through a cross-sectional area (remarks: maximum two insulated objects in series)

c) Analyse graphs of temperature-distance (T-L) for heat conduction through insulated and non-insulated rods.
   **maximum two rods in series*

8.4 Thermal expansion

a) Define coefficient of linear expansion, α, area expansion, β and volume expansion, γ

b) Solve problems related to thermal expansion of linear, area and volume.
   **include expansion of liquid in a container*
   $\Delta L = \alpha L_o \Delta T$, $\Delta A = \beta A_o \Delta T$, $\Delta V = \gamma V_o \Delta T$, $\beta = 2\alpha$, $\gamma = 3\alpha$

Topic 9: Kinetic Theory of Gases and Thermodynamics

9.1 Kinetic theory of gases
   a) State the assumptions of kinetic theory of gases.
   b) Describe root mean square (rms) speed of gas molecules, $v_{rms} = \sqrt{\langle v^2 \rangle}$
   c) Solve problems related to root mean square (rms) speed of gas molecules,
      $v_{rms} = \sqrt{\dfrac{3kT}{m}} = \sqrt{\dfrac{3RT}{M}}$
   d) Solve problems related to the equations, $PV = \frac{1}{3}Nm{v_{rms}}^2$ and pressure , $P = \frac{1}{3}\rho {v_{rms}}^2$

9.2 Molecular kinetic energy and internal energy
   a) Explain and use translational kinetic energy of a molecule, $K_{tr} = \frac{3}{2}\left(\frac{R}{N_A}\right)T = \frac{3}{2}kT$
   b) Define degree of freedom.
   c) Identify number of degrees of freedom, f for monoatomic, diatomic and polyatomic gas molecules.
   d) State the principle of equipartition of energy.
   e) Discuss internal energy of gas.
   f) Solve problems related to internal energy, $U = \frac{1}{2}fNkT$

9.3 First Law of Thermodynamics
   a) State the First Law of Thermodynamics, $\Delta U = Q - W$
   b) Solve problem related to First Law of Thermodynamics.

9.4 Thermodynamic processes
   a) Define the following thermodynamic processes:
      i. Isothermal;
      ii. Isochoric;
      iii. Isobaric and
      iv. Adiabatic.
   b) Analyse P-V graph for all the thermodynamic processes.

9.5 Thermodynamics work
   a) Derive equation of work done in isothermal, isochoric and isobaric processes from P-V graph.
   b) Solve problem related to work done in:
      i. isothermal process, $W = nRT \ln\frac{V_f}{V_i} = nRT \ln\frac{P_i}{P_f}$
      ii. isobaric process, $W = \int P\mathrm{d}V = P(V_f - V_i)$
      iii. isochoric process, $W = \int P\mathrm{d}V = 0$

Panels:

No.	Panel	Institution	Academic Qualification
1.	Assoc. Professor Ts. Dr. Shahrul Kadri bin Ayop	Universiti Pendidikan Sultan Idris	PhD (Electronics for Informatics), Hokkaido MSc (Physics), Leipzig BSc (Hons) (Industrial Physics), UTM
2.	Prof. Ts. Dr. Azhan bin Hashim @ Ismail	Universiti Teknologi MARA	PhD (Material Science), UPM BSc (Hons) – Physics, UPM
3.	Dr. Ariffin bin Abas	Kolej Matrikulasi Johor	PhD(Physics), UPM MSc (Physics), UPM BSc (Hons) – Physics, UM DipEd (Physics), UM
4.	Pn. Salmah Binti Othman	Kolej Matrikulasi Melaka	BSc (Hons) – Physics, UTM DipEd (Physics), UTM
5.	En Marhalim bin Hashim	Kolej Matrikulasi Pulau Pinang	MMngtTech (HRD) UTM, BSc Ed (Hons) - Physics, UKM
6.	En Mohd Fauzi bin Abdul Zubir	Kolej Matrikulasi Sarawak	BSc (Hons) – Fizik dan Matematik, UM Diploma Pendidikan (Pengajian Fizik) MPRM
7.	Pn. Raziah binti Mohd Noor	Kolej Matrikulasi Kejuruteraan Kedah	B.EE (Hons) – Power
8.	Tn. Hj. Zahidi bin Hashim	Kolej Matrikulasi Kejuruteraan Johor	BSc with Ed. (Hons) – Physics, UTM
9.	Pn. Chong Yoke Lai	Kolej Matrikulasi Johor	BSc Comp. with Ed. (Hons) – Physics, UTM
10.	En Mohd Rohit bin Safuan	Kolej Matrikulasi Selangor	BSc Ed. (Hons) – Physics, USM
11.	Pn. Siti Aisyah binti Sahdan	Bahagian Pembangunan Kurikulum KPM	BSc. Physics (Hons), University of Nottingham, UK
12.	Pn. Aifaa binti Awang Kechik	Bahagian Matrikulasi KPM	BSc with Ed. (Hons) – Physics, UPM
13.	Pn. Jamilah binti Abdul Rashid	Bahagian Matrikulasi KPM	BEng (Hons) Electrical, Electronis & System, UKM